
Low-Rank Neural Networks at the Edge of Convexity: Finite-Width NTK Feynman Diagrams

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Abstract

1 Low-rank neural networks are usually sold as smaller models. This paper asks
2 a sharper question: when does low rank still preserve the optimization geometry
3 that makes wide networks trainable? We answer this through the finite-width
4 neural tangent kernel. Our main result is that low-rank training is governed by a
5 genuine three-way compromise between width, depth, and rank. To the best of our
6 knowledge, this is the first NTK analysis showing that an optimization certificate
7 can impose a depth-rank tradeoff of this form. In particular, for low-rank bottleneck
8 dynamics, preserving the NTK spectral margin requires the rank to grow cubically
9 with depth, $r \gtrsim L^3$, up to dataset-size and logarithmic factors. This cubic depth law
10 could explain why very deep low-rank networks become difficult to train efficiently:
11 depth increases the rank needed to preserve a stable optimization geometry. We
12 also give a full-rank-compatible parametrization, separating true low-rank effects
13 from ordinary finite-width effects. True finite-network NTK experiments confirm
14 the exact full-rank matching, the predicted operator scaling, the rank-depth tradeoff,
15 and the finite-network cumulant growth. The result is a criterion for when low-rank
16 networks remain trainable for structural reasons, not merely parameter-efficient
17 ones.

18 1 Introduction

19 Low-rank neural networks are usually motivated by efficiency: fewer parameters, lower memory
20 traffic, and cheaper adaptation. This motivation is incomplete. Rank also changes the geometry of
21 training. In the NTK regime, gradient descent is close to kernel regression when the empirical NTK
22 Gram matrix is positive and stable. Thus a low-rank network is useful not only when it is small,
23 but when its finite NTK stays close enough to a positive limiting kernel. We call this the edge of
24 convexity: the regime where the network is finite and structured, but the NTK Gram matrix still
25 provides a convex local training geometry.

26 This question is difficult because three limits interact. Width controls ordinary finite-width fluctua-
27 tions, depth amplifies NTK tensor corrections, and rank changes the channel contractions themselves.
28 Treating low rank as merely replacing $1/n$ by $1/r$ misses the full-rank endpoint: if a rank- n fac-
29 torization is just an orthogonal change of variables, the low-rank correction must vanish exactly.
30 Conversely, bottleneck random-feature low-rank networks do not reduce to dense MLPs and can
31 suppress old NTK paths geometrically. A useful theory must separate these two cases.

32 We study scalar-output ReLU networks with layer weights of the form

$$W^{(\ell)} = \frac{\sigma_w}{\sqrt{n_{\ell-1}}} \sqrt{\frac{n_{\ell}}{r_{\ell}}} U^{(\ell)} B^{(\ell)}, \quad (U^{(\ell)})^{\top} U^{(\ell)} = I_{r_{\ell}}, \quad (1)$$

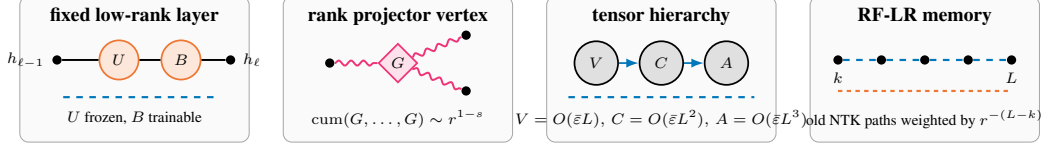


Figure 1: Main diagrammatic objects, drawn in the color and line vocabulary used throughout the finite-width NTK diagram literature. Solid black lines denote NNGP/preactivation transport, dashed colored lines denote NTK legs, orange and pink vertices denote rank and Haar-projector cumulants, and gray blobs denote transported tensor cumulants.

where $U^{(\ell)}$ is frozen Haar-Stiefel and $B^{(\ell)}$ is trainable. Haar-Stiefel means that the columns of $U^{(\ell)}$ form an orthonormal r_ℓ -frame in $\mathbb{R}^{n_\ell}$, chosen uniformly among all such frames. Equivalently, $U^{(\ell)}$ selects a random r_ℓ -dimensional subspace with no preferred direction. When $r_\ell = n_\ell$, the map $B^{(\ell)} \mapsto U^{(\ell)} B^{(\ell)}$ is an orthogonal reparametrization of a dense MLP. Thus the full-rank endpoint is exact, not asymptotic. Away from full rank, the relevant random object is the normalized projector

$$G = \frac{n}{r} U U^\top, \quad (2)$$

whose covariance scale is

$$\gamma_{n,r} = \frac{n(n-r)}{r(n-1)(n+2)} = \frac{1}{r} - \frac{1}{n} + O(n^{-2}). \quad (3)$$

The mixed perturbation parameter is therefore $1/n + \gamma_{n,r}$. The term $1/n$ is the ordinary dense finite-width correction, while $\gamma_{n,r}$ is the rank defect. The latter vanishes at $r = n$, as required by exact full-rank compatibility.

This MLP-compatible model should not be confused with a random-feature low-rank bottleneck (RF-LR). In that regime, the limiting NTK recursion contains an explicit factor $1/r$ on old NTK lines. At ReLU edge of chaos, old memory from layer k to layer L is suppressed as $(2r)^{-(L-k)}$. This creates a different spectral design rule: low rank shortens memory, but the centered deterministic gap is also smaller. The paper is organized around this separation.

Contributions.

- We prove exact full-rank matching: at $r = n$, the factorized network and dense MLP have identical empirical NTKs pathwise.
- We identify the low-rank Feynman vertices as connected cumulants of $G = (n/r) U U^\top$, with second cumulant scale $\gamma_{n,r}$.
- Under explicit ReLU transport assumptions, we derive tensor scalings $V = O(\varepsilon L)$, $C = O(\varepsilon L^2)$, and $A = O(\varepsilon L^3)$, with endpoint constants 5, 21/20, and 173/720 for isolated off-diagonal modes.
- Under an explicit well-spread data assumption, we prove operator concentration for finite empirical NTKs at scale $L^{3/2} \sqrt{m \varepsilon} + mL^3 \varepsilon$, up to logarithmic factors, and show that dNTK/ddNTK descendants introduce no new random vertices.
- We derive a conservative rank-depth stability rule for bottleneck low-rank dynamics: to preserve the centered NTK spectral margin in the unprojected RF-LR model, rank scales cubically with depth, $r \gtrsim mL^3$, up to logarithmic factors. We also state the separate centered/angularized mL^2 rule with its additional projection hypothesis.
- We validate the theory with proxy experiments and true finite-network autodiff NTKs, including full-rank matching, rank-defect scaling, operator ratios, and direct cumulant growth.

Status of the claims. The paper separates what is proved algebraically from what is conditional on standard concentration hypotheses. Full-rank matching, the fixed-left gradient decomposition, the Haar covariance, and the scalar endpoint moments are unconditional calculations. The depth hierarchy and operator bounds are stated under the ReLU edge-of-chaos transport and well-spread data assumptions made explicit below. Direct finite-network cumulants are treated as diagnostics of the polynomial hierarchy, not as direct measurements of isolated Feynman coefficients.

2 Related Work

The NTK was introduced as the infinite-width kernel governing gradient-flow training [1]. This line of work changed the community’s view of overparameterized networks: in the infinite-width lazy regime, training can be studied through a deterministic kernel, and convergence depends on the spectrum of the NTK Gram matrix. Follow-up work showed that wide networks of any depth evolve approximately as linear models around initialization [2], and that overparameterized networks can be trained globally under width conditions that keep the empirical NTK stable [3, 4, 5]. These results explain why positive NTK spectra are powerful, but they mostly treat dense networks and do not describe the finite rank-width corrections introduced by factorized layers.

A second thread studies the limits and finite-width corrections of the NTK. Tensor programs give a general language for infinite-width limits and parametrization choices across architectures [8, 9]. The neural tangent hierarchy and related expansions describe how finite-width dynamics depart from the frozen-kernel limit [10]. Feynman-diagram methods give a complementary perturbative calculus for wide networks, organizing corrections by connected cumulants and tensor contractions [11, 12]. Our work follows this finite-width viewpoint, but replaces dense-channel contractions by cumulants of the Haar projector induced by low rank.

The closest optimization and spectral guarantees are complementary rather than substitutive. Prior work proves global convergence for deep networks with one wide layer followed by a pyramidal topology, showing that not every layer must be wide [6]. Tight lower bounds are also known for the smallest eigenvalue of deep ReLU NTKs, including finite-width architectures with one layer of order $\tilde{\Omega}(m)$ and other widths essentially arbitrary [7]. These results already cover strong “one wide layer,” bottleneck-width, global-convergence, and $\lambda_{\min}$ -type guarantees for full-rank networks. Our claim is different: we do not try to improve their global convergence conditions. We provide a finite-rank perturbation theory for factorized layers $W = \sqrt{n/r}UB$, isolate the exact rank-defect parameter $\gamma_{n,r}$, prove exact full-rank matching at $r = n$, and convert the finite-rank NTK deviation into a Weyl spectral-preservation criterion. Recent quantitative GP approximation during training [16] is also orthogonal: it controls Wasserstein distance to a Gaussian-process limit for trained shallow networks, whereas our object is operator concentration of $\hat{\Theta} - K_*$ for deep low-rank NTKs and the resulting stability of the centered spectral margin.

A third thread concerns practical NTK computation and low-rank structure. Empirical NTKs can be computed efficiently enough to probe finite networks directly [14]. Low-rank adaptation and low-rank training have also been analyzed in NTK-like regimes, including results showing that sufficiently large low-rank adapters avoid bad local minima in certain settings [15]. These works motivate low rank as an optimization and adaptation tool. Our contribution is different: we ask when low rank preserves the NTK convexity certificate at finite width and finite depth, and we show that the answer depends on whether the rank constraint is an isometric perturbation of the dense MLP or a genuine RF-LR bottleneck.

3 Models and Full-Rank Matching

Let $x_1, \dots, x_m$ be a fixed dataset. The empirical scalar-output NTK is

$$\hat{\Theta}_{ab} = \nabla_{\theta} f(x_a) \cdot \nabla_{\theta} f(x_b). \quad (4)$$

For the factorized model (1), trainable parameters are the right factors $B^{(\ell)}$ and the readout. We deliberately keep the left Stiefel factors $U^{(\ell)}$ frozen. This is not only a computational choice. It is what makes the low-rank model a controlled NTK perturbation: the trainable tangent space is a fixed linear subspace, and all rank-dependent randomness enters through the static projector $G = (n/r)UU^{\top}$. If U is trained as well, the tangent space itself moves, producing additional dNTK terms that couple variations of U and B .

Theorem 3.1 (Pathwise full-rank matching). *If $r_{\ell} = n_{\ell}$ for every hidden layer, then for every realization, every dataset, and every depth, the empirical NTK in B -coordinates equals the empirical NTK of the dense MLP in dense-weight coordinates $\widetilde{W}^{(\ell)} = U^{(\ell)}B^{(\ell)}$. The same equality holds for the NNGP, dNTK, ddNTK, and all tensor cumulants built from them.*

119 *Proof.* When $r = n$, U is orthogonal. The linear map $\Delta B \mapsto U\Delta B$ preserves Frobenius inner
120 products. The Jacobian with respect to B is the dense-weight Jacobian composed with this isometry;
121 hence $J_B J_B^\top = J_W J_W^\top$. Higher derivatives and cumulants are invariant under the same coordinate
122 change. $\square$

123 **Proposition 3.2** (Why the left factor is frozen). *Consider one layer $W = \alpha U B$, with $U^\top U = I_r$,
124 $\alpha > 0$, and loss gradient $H = \partial \mathcal{L} / \partial W$. If only B is trained, then*

$$\nabla_B \mathcal{L} = \alpha U^\top H, \quad \dot{W}_B = -\eta \alpha^2 U U^\top H. \quad (5)$$

125 *The layer motion is the fixed projected gradient through $U U^\top$. If U is also trained on the Stiefel
126 manifold, then the projected Riemannian gradient contributes*

$$\dot{W}_U = -\eta \alpha^2 (I - U U^\top) H B^\top B \text{ up to the skew-gauge component inside } \text{span}(U). \quad (6)$$

127 *Consequently the empirical NTK and its time derivative contain additional U - B interaction blocks,
128 proportional to contractions with $B^\top B$. These blocks are absent in the frozen-left model and are not
129 controlled solely by Haar-projector cumulants of $G = (n/r) U U^\top$.*

130 *Proof.* For fixed U , the differential is $dW = \alpha U dB$. Hence $d\mathcal{L} = \langle H, \alpha U dB \rangle = \langle \alpha U^\top H, dB \rangle$,
131 giving (5). If B is fixed and U varies on the Stiefel manifold, then $dW = \alpha dU B$ and the Euclidean
132 gradient is $\alpha H B^\top$. Projecting this gradient to the normal component of the Stiefel tangent gives
133 $(I - U U^\top) \alpha H B^\top$, while tangent rotations inside $\text{span}(U)$ are gauge-coupled with B . Multiplying
134 the resulting dU by B gives the displayed normal contribution (6). Thus training U introduces terms
135 depending on both H and $B^\top B$, so the tangent kernel contains $B B^\top$ -weighted and cross U/B blocks
136 in addition to the fixed-projector B -block. Differentiating the NTK along gradient flow then creates
137 dNTK terms involving $\dot{U}$, $\dot{B}$, and products such as $B^\top B$. These are precisely the moving-subspace
138 interactions excluded by freezing U . $\square$

139 Proposition 3.2 explains the modeling choice. Training both factors may be useful in practice, but it
140 is a different dynamical system: the low-rank subspace rotates during training, and the NTK drift
141 receives extra moving-projector terms. Freezing U isolates the rank-width effect we can prove and
142 keeps the diagrammatics closed under the Haar projector vertices of Lemma 4.1.

143 The RF-LR bottleneck has a different recursion,

$$\Theta^{(\ell)} = 1 + \frac{1}{r} \Theta^{(\ell-1)} \dot{\Sigma}^{(\ell)} + \frac{1}{r} \Sigma^{(\ell)}. \quad (7)$$

144 At the ReLU fixed point $\dot{\Sigma}^{(\ell)} \rightarrow 1/2$, so an old NTK contribution is multiplied by approximately
145 $(2r)^{-(L-k)}$. This is a different model, not a perturbation of the dense MLP.

146 4 Haar Projector Vertices

147 Let $U \in \mathbb{R}^{n \times r}$ be Haar-Stiefel and $G = (n/r) U U^\top$. Orthogonal invariance gives the full covariance
148 tensor.

149 **Lemma 4.1** (Haar projector covariance). *For all indices,*

$$\text{Cov}(G_{ij}, G_{kl}) = \gamma_{n,r} \left(\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk} - \frac{2}{n} \delta_{ij} \delta_{kl} \right), \quad \gamma_{n,r} = \frac{n(n-r)}{r(n-1)(n+2)}. \quad (8)$$

150 *In particular $\gamma_{n,n} = 0$.*

151 *Proof.* The covariance of a Haar projector has the invariant form $a(\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk}) + b \delta_{ij} \delta_{kl}$. The
152 identity $\text{tr } G = n$ gives $2a + nb = 0$. The standard second moment of a Haar projector gives
153 $a = \gamma_{n,r}$, hence (8). $\square$

154 Higher cumulants are given by orthogonal Weingarten contractions. Connected s -point cumulants
155 scale as $O_s(r^{1-s})$ and vanish in the full-rank endpoint. Thus the low-rank Feynman rule is simple:
156 dense Gaussian-ReLU vertices remain, and each low-rank correction inserts connected cumulants of
157 G .

158 **Lemma 4.2** (Rank cumulant power counting). *Let Z_a , $a = 1, \dots, r$, be iid rank-channel atoms with*
 159 *uniformly bounded moments of all orders, and define empirical observables*

$$X_i^{(r)} = r^{-a_i} \sum_{a=1}^r f_i(Z_a), \quad i = 1, \dots, s.$$

160 *Then the connected cumulant satisfies*

$$\text{cum}\left(X_1^{(r)}, \dots, X_s^{(r)}\right) = r^{1-\sum_i a_i} \text{cum}\left(f_1(Z), \dots, f_s(Z)\right). \quad (9)$$

161 *In particular, normalized empirical averages have s -point cumulants of order r^{1-s} .*

162 *Proof.* By multilinearity of cumulants,

$$\text{cum}(X_1^{(r)}, \dots, X_s^{(r)}) = r^{-\sum_i a_i} \sum_{a_1, \dots, a_s=1}^r \text{cum}(f_1(Z_{a_1}), \dots, f_s(Z_{a_s})).$$

163 Independence kills every term whose indices are not all in the same connected block. Only $a_1 =$
 164 $\dots = a_s$ contributes, giving r identical terms and hence (9). $\square$

165 This elementary lemma is the right-factor-only analogue of the usual width power counting. It
 166 explains why finite-rank bias terms are typically $O(r^{-1})$ after averaging over seeds, while single-seed
 167 fluctuations are $O_{\mathbb{P}}(r^{-1/2})$. It also explains why the exact full-rank endpoint must be treated with the
 168 projector defect $G - I$, rather than by the informal replacement $n \mapsto r$.

169 5 Tensor Scaling at ReLU Edge of Chaos

170 For normalized ReLU, the edge-of-chaos correlation asymptotics used in the RF-LR analysis [13]
 171 give $\theta_\ell = 3\pi/\ell + O(\log \ell/\ell^2)$, and

$$\dot{C}(\rho_\ell) = 1 - \frac{3}{\ell} + O\left(\frac{\log \ell}{\ell^2}\right). \quad (10)$$

172 Transporting an external NTK leg from layer k to L gives the factor $(k/L)^\lambda$ for an exponent
 173 $\lambda \in \{0, 3\}$, depending on whether the Gram entry is diagonal or off-diagonal.

174 **Lemma 5.1** (Transport-source summation). *Let $T_\ell \in \mathbb{R}^d$ satisfy*

$$T_{\ell+1} = \left(I - \frac{M}{\ell} + E_\ell\right) T_\ell + \ell^\beta (s + e_\ell), \quad (11)$$

175 *where $\|E_\ell\| \leq C \log^p(\ell)/\ell^2$, $\|e_\ell\| \leq C \log^q(\ell)/\ell$, and $M + (\beta + 1)I$ is invertible. Then*

$$T_L = L^{\beta+1} (M + (\beta + 1)I)^{-1} s + O(L^\beta \log^{p+q+2} L). \quad (12)$$

176 *Proof.* Iterating (11) gives a variation-of-constants formula. The product of propagators from k to L
 177 is $(k/L)^M$ up to a multiplicative error whose accumulated contribution is $O(L^\beta \log^{p+q+2} L)$, since
 178 $\sum_k \log^p(k)/k^2 < \infty$. Thus

$$T_L = L^{-M} \sum_{k \leq L} k^{M+\beta I} s + O(L^\beta \log^{p+q+2} L).$$

179 Euler-Maclaurin yields

$$L^{-M} \sum_{k \leq L} k^{M+\beta I} = L^{\beta+1} \int_0^1 u^{M+\beta I} du + O(L^\beta),$$

180 and the matrix integral is $(M + (\beta + 1)I)^{-1}$. $\square$

181 Let $\bar{\varepsilon} = \max_{\ell \leq L} (1/n_\ell + \gamma_{n_\ell, r_\ell})$. The finite-width tensors obey

$$V_L = O(\bar{\varepsilon}L), \quad C_L = O(\bar{\varepsilon}L^2), \quad A_L = O(\bar{\varepsilon}L^3), \quad (13)$$

182 where C denotes the mixed NNGP-NTK block (D, F) , and A denotes the NTK-NTK block (A, B) .

183 **Proposition 5.2** (Conditional tensor hierarchy). *Assume the low-rank cumulant recursion closes on*
 184 *the same tensor sectors as the dense finite-width Feynman hierarchy and that the ReLU edge-of-chaos*
 185 *transport is uniform away from the diagonal singularity. Then the isolated tensor coordinates satisfy*

$$V_L = O(\bar{\varepsilon}L), \quad D_L, F_L = O(\bar{\varepsilon}L^2), \quad A_L, B_L = O(\bar{\varepsilon}L^3). \quad (14)$$

186 Consequently, the perturbative mean-NTK correction is controlled in the regime $L^3\bar{\varepsilon} \ll 1$.

187 *Proof.* The fresh NNGP four-point source contributes $O(\bar{\varepsilon})$ per layer and has no transported NTK
 188 leg in the dominant radial mode, so summation gives $V_L = O(\bar{\varepsilon}L)$. The mixed blocks D, F have one
 189 transported NTK leg and therefore one transport-source summation with $\beta = 1$, yielding $O(\bar{\varepsilon}L^2)$
 190 by Lemma 5.1. The NTK-NTK blocks A, B have two transported NTK legs, equivalently source
 191 exponent $\beta = 2$, giving $O(\bar{\varepsilon}L^3)$. These are isolated tensor coordinates; scalar-output cumulants can
 192 mix these sectors. $\square$

193 **Endpoint constants.** At the ReLU endpoint, set $X_0 = 2(g_+)^2$ and $Y_0 = 2\mathbf{1}_{\{g>0\}}$. Then

$$\text{Var}(X_0) = 5, \quad \text{Cov}(X_0, Y_0) = 1, \quad \text{Var}(Y_0) = 1. \quad (15)$$

194 For a transported off-diagonal leg, $\lambda = 3$. The mixed and NTK-NTK endpoint integrals are

$$B_\lambda = \frac{1}{\lambda+1} \left(5 - \frac{4}{\lambda+2} \right), \quad A_{\lambda,\mu} = \frac{1}{(\lambda+1)(\mu+1)} \left(5 - \frac{4}{\lambda+2} - \frac{4}{\mu+2} + \frac{4}{\lambda+\mu+3} \right). \quad (16)$$

195 Thus for off-diagonal modes,

$$V : 5, \quad C : B_3 = \frac{21}{20}, \quad A : A_{3,3} = \frac{173}{720}. \quad (17)$$

196 These constants are tensor coefficients after pairing-basis extraction, not arbitrary scalar-output cumu-
 197 lants. Direct finite-network cumulants mix Wick/Wishart terms, radial modes, readout contractions,
 198 and several Feynman sectors; they are therefore expected to show finite-depth effective slopes rather
 199 than the isolated constants exactly.

200 5.1 A three-layer finite-rank calculation

201 The simplest place where the low-rank simplification is visible is a three-layer network, i.e. two
 202 ReLU feature maps. Let

$$Q_x = \frac{1}{r} \sum_{a=1}^r X_a^2, \quad Q_y = \frac{1}{r} \sum_{a=1}^r Y_a^2, \quad S = \frac{1}{r} \sum_{a=1}^r X_a Y_a, \quad C_r = \frac{S}{\sqrt{Q_x Q_y}}.$$

203 For the right-factor-only low-rank model, the empirical second hidden-layer kernel has the schematic
 204 form

$$K_{2,r} = \sqrt{Q_x Q_y} \kappa(C_r), \quad \dot{K}_{2,r} = \dot{\kappa}(C_r), \quad \hat{\Theta}_r^{(3)} = K_{2,r} + S \dot{K}_{2,r},$$

205 where κ is the normalized ReLU covariance map.

206 **Proposition 5.3** (Delta-method correction for three layers). *For fixed correlation $\rho \in (-1, 1)$,*
 207 *there is an explicit smooth function $\Delta^{(3)}(\rho)$, depending only on $\kappa, \dot{\kappa}$ and the covariance matrix of*
 208 *(X^2, Y^2, XY) , such that*

$$\mathbb{E} \hat{\Theta}_r^{(3)}(\rho) = \Theta_\infty^{(3)}(\rho) + \frac{1}{r} \Delta^{(3)}(\rho) + O(r^{-2}), \quad \hat{\Theta}_r^{(3)} - \mathbb{E} \hat{\Theta}_r^{(3)} = O_{\mathbb{P}}(r^{-1/2}). \quad (18)$$

209 For scale-invariant ReLU on the diagonal, $\Delta^{(3)}(1) = 0$.

210 *Proof.* Write $\widehat{\Theta}_r^{(3)} = F(Q_x, Q_y, S)$. By the multivariate central limit theorem and Lemma 4.2,

$$(Q_x, Q_y, S) = (1, 1, \rho) + r^{-1/2}\xi + O_{\mathbb{P}}(r^{-1})$$

211 with ξ asymptotically Gaussian and covariance determined by the moments of (X^2, Y^2, XY) . Taylor
212 expansion gives

$$\mathbb{E}F(Q_x, Q_y, S) = F(1, 1, \rho) + \frac{1}{2r} \text{tr}(\nabla^2 F(1, 1, \rho) \Sigma_\rho) + O(r^{-2}),$$

213 which is (18). The $O_{\mathbb{P}}(r^{-1/2})$ fluctuation is the first-order delta-method term. On the diagonal
214 $X = Y$, ReLU scale invariance makes the normalized correlation and derivative atoms deterministic
215 along the radial direction, cancelling the first mean correction. $\square$

216 6 Operator Concentration and Spectral Preservation

217 Let $\Theta_{\infty, L}$ be the deterministic limiting NTK and $\widehat{\Theta}_L$ the empirical finite NTK. Entrywise tensor
218 bounds give

$$\text{Var}((\widehat{\Theta}_L)_{ab}) \lesssim L^3 \bar{\varepsilon}, \quad |\mathbb{E}(\widehat{\Theta}_L)_{ab} - (\Theta_{\infty, L})_{ab}| \lesssim L^3 \bar{\varepsilon}. \quad (19)$$

219 A layer martingale and matrix Freedman yield the operator form. The well-spread data assumption
220 below means that no small group of examples dominates the random feature directions; equivalently,
221 the matrix variance of the empirical NTK fluctuation grows like m , not m^2 .

222 **Theorem 6.1** (Finite-network operator concentration). *Assume $L^3 \bar{\varepsilon} \leq c$. Assume further that the*
223 *data are well-spread in the layerwise feature bases, so the entrywise tensor variance bounds upgrade*
224 *to a matrix variance proxy of order $mL^3 \bar{\varepsilon}$. With probability at least $1 - \delta$,*

$$\|\widehat{\Theta}_L - \Theta_{\infty, L}\|_{\text{op}} \leq CL^{3/2} \sqrt{m \bar{\varepsilon}} \text{polylog}(m, L, 1/\delta) + CmL^3 \bar{\varepsilon} \text{polylog}(m, L, 1/\delta). \quad (20)$$

225 *A robust version replaces the first term by $CmL^{3/2} \sqrt{\bar{\varepsilon}}$ without the well-spread data assumption.*

226 *Proof sketch.* Expose the random weights and projectors layer by layer and decompose $\widehat{\Theta}_L - \mathbb{E}\widehat{\Theta}_L$ as
227 a matrix martingale. Proposition 5.2 gives conditional entrywise variance $O(L^3 \bar{\varepsilon})$ and conditional bias
228 $O(L^3 \bar{\varepsilon})$. Under the well-spread data assumption, the predictable quadratic variation is $O(mL^3 \bar{\varepsilon})$ in
229 operator norm rather than $O(m^2 L^3 \bar{\varepsilon})$. Matrix Freedman gives the first term in (20), with logarithmic
230 losses from truncation and union over tensor coordinates. The deterministic bias contributes the
231 second term. Without this assumption, the same argument uses the Frobenius-to-operator bound on
232 the variance proxy and loses a factor $\sqrt{m}$. $\square$

233 If $\mu_\infty(L) = \lambda_{\min}(P_m \Theta_{\infty, L} P_m)$ is the limiting centered spectral margin, Weyl's inequality gives
234 preservation whenever the right-hand side of (20) is at most $\mu_\infty(L)/2$. In the RF-LR bottleneck, the
235 baseline is not the dense MLP margin; the centered proxy margin can scale as $1/(rL)$. Combining
236 this with the bottleneck deviation gives the conservative design rule $r \gtrsim mL^3$, up to logarithmic
237 factors, when the RF-LR perturbation scale is $1/r$.

238 6.1 RF-LR memory and rank-depth rules

239 The RF-LR recursion (7) is not a full-rank-compatible perturbation. Its old-kernel memory obeys an
240 exact product rule:

$$\Theta_{\text{old} \leftarrow k}^{(L)} = \Theta^{(k)} \prod_{j=k+1}^L \frac{\dot{\Sigma}^{(j)}}{r}. \quad (21)$$

241 At the normalized ReLU edge of chaos, $\dot{\Sigma}^{(j)} = 1/2 + O(j^{-1})$, hence

$$\left| \Theta_{\text{old} \leftarrow k}^{(L)} \right| \leq C \left(\frac{1}{2r} \right)^{L-k} \left(\frac{L}{k} \right)^C. \quad (22)$$

242 Low rank therefore shortens old NTK memory geometrically. The cost is that the centered determin-
243 istic margin can shrink with r and L .

Proposition 6.2 (Unprojected and centered RF-LR stability rules). *Consider an RF-LR bottleneck whose centered limiting margin satisfies $\mu_{\text{RF}}(L, r) \gtrsim (rL)^{-1}$. If the empirical operator deviation is controlled by the conservative scale $L^{3/2}\sqrt{m/r} + mL^3/r$, then spectral preservation follows from*

$$r \gtrsim mL^3 \text{polylog}(m, L, 1/\delta). \quad (23)$$

If the radial mode is projected out and an angularized centered decomposition improves the variance scale by one power of L , then the sufficient rule improves to

$$r \gtrsim mL^2 \text{polylog}(m, L, 1/\delta). \quad (24)$$

Proof. Weyl’s inequality requires the operator deviation to be at most a fixed fraction of $\mu_{\text{RF}}(L, r)$. In the unprojected RF-LR model, the radial contribution remains present in the A -sector, whose tensor scale is cubic in depth. The sufficient inequality is therefore dominated by the mL^3/r contribution after absorbing logarithms and constants, giving (23). In the centered/angularized model, the radial sector is removed before concentration and the dominant tensor variance is quadratic in depth. Repeating the same Weyl argument gives (24). The second statement is conditional on this projection and should not be read as the default RF-LR rule. $\square$

7 NTK Descendants and Training-Time Drift

Let $J_c = \nabla_{\theta} f(x_c)$ and define the dNTK and ddNTK descendants

$$\Xi_{ab;c} = D\Theta_{ab}[J_c], \quad \Psi_{ab;cd} = D^2\Theta_{ab}[J_c, J_d]. \quad (25)$$

These descendants add derivative legs to the same layer maps. They do not introduce new independent random variables.

Proposition 7.1 (No new low-rank vertices). *The joint cumulants of (K, Θ, Ξ, Ψ) are generated by the same Gaussian-ReLU atoms and Haar-projector cumulants of $G = (n/r)UU^{\top}$. Differentiation changes deterministic contractions but not the probabilistic vertex set.*

Proof. Each derivative of the network output with respect to a trainable right factor $B^{(\ell)}$ inserts an additional deterministic backpropagated leg and a factor of the same frozen projector $UU^{\top}$. Higher derivatives differentiate ReLU gates and covariance maps, changing the local deterministic tensors $\kappa, \dot{\kappa}, \ddot{\kappa}$, but all randomness still comes from the original Gaussian preactivations, readout weights, and frozen projectors. Since U is fixed, there is no moving-projector derivative. Thus the connected diagrams for K, Θ, Ξ, Ψ have the same random vertices as the NTK diagrams and only differ in their deterministic leg labels. $\square$

Under square-loss gradient flow, $\dot{\Theta}_{ab,t} = -\sum_c \Xi_{ab;c,t} e_{c,t}$. The descendant concentration bound therefore yields, in a lazy window,

$$\|\Theta_t - \Theta_0\|_{\text{op}} \leq Ct \|e_0\|_2 \left(L^3 \sqrt{m\bar{\varepsilon}} + mL^3 \bar{\varepsilon} \right) \text{polylog}(m, L, 1/\delta). \quad (26)$$

8 Experiments

We run three classes of experiments and present the figures in the same order as the theory. Proxy experiments first validate the rank-projector law and operator normalization. We then check the endpoint tensor constants and RF-LR memory suppression. Finally, true finite-network experiments compute empirical NTKs by PyTorch autograd and estimate direct finite-network covariances of the NNGP Gram and empirical NTK.

9 Limitations

The theory is for signed/isometric low-rank factors and RF-LR bottlenecks, not strict nonnegative matrix factorization. The constants 5, 21/20, and 173/720 are isolated proxy/tensor constants, not direct scalar-output cumulant constants. The operator concentration theorem keeps logarithmic factors and a well-spread data distinction; removing these losses is open. A complete numerical table for all dNTK/ddNTK descendant coefficients is left for future work.

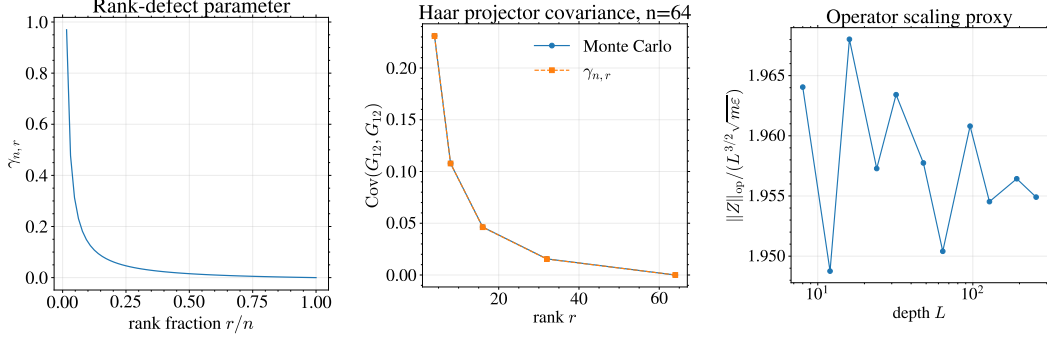


Figure 2: Rank-projector and operator checks. Left: the rank-defect scale $\gamma_{n,r}$ vanishes at full rank. Middle: empirical Haar-projector covariance matches the closed form. Right: the normalized operator proxy $\|Z\|_{\text{op}}/(L^{3/2}\sqrt{m\epsilon})$ is approximately stable across depth.

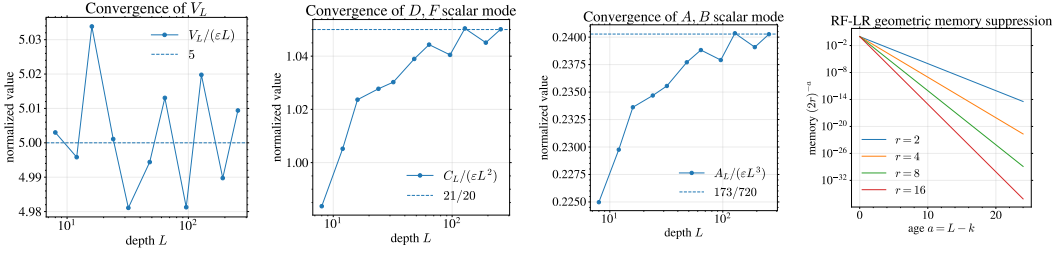


Figure 3: Tensor constants and RF-LR memory. The first three panels show convergence to the isolated constants 5, 21/20, and 173/720. The last panel shows geometric suppression of old NTK memory in the RF-LR bottleneck.

10 Conclusion

Low rank changes finite-width NTK theory not only by reducing parameters, but by changing the stability threshold for training. In the unprojected RF-LR bottleneck, old NTK memory is geometrically shortened, but the centered spectral margin is also smaller. Combining these effects gives the main design message: preserving the NTK edge of convexity has a conservative cubic-depth sufficient rule, $r \gtrsim mL^3$, up to logarithmic factors. A quadratic-depth rule is available only after the radial mode is projected out. The rank-width diagrammatics developed here connect these criteria to conditional operator concentration and to finite-network experiments.

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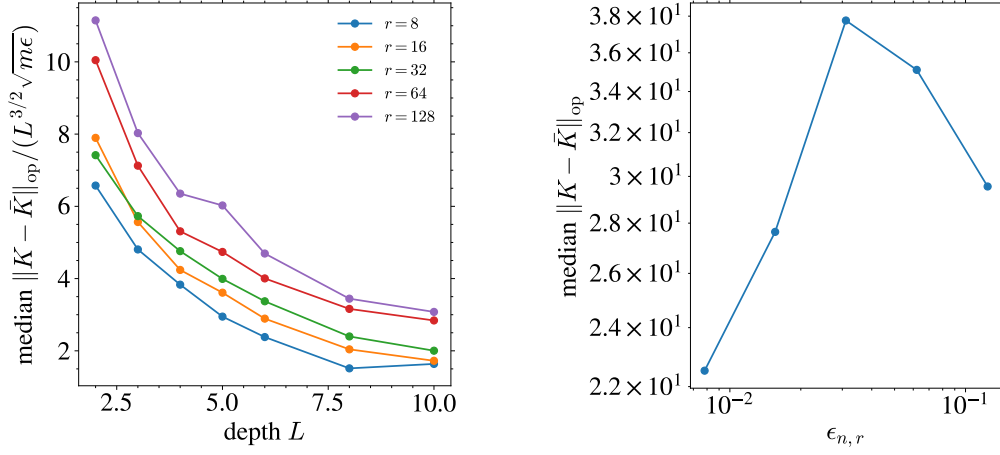


Figure 4: True finite-network NTK checks computed by autodiff. Left: empirical operator deviations follow the predicted normalization. Right: the finite-network deviation tracks the rank-defect scale and vanishes at the full-rank endpoint.

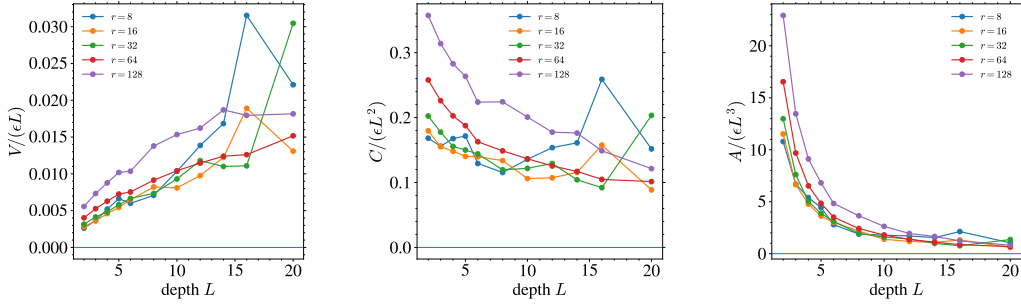


Figure 5: True finite-network cumulants from autodiff NTKs. These scalar-output cumulants are not isolated Feynman coefficients; they test the predicted polynomial control and rank-width scaling on actual finite networks.

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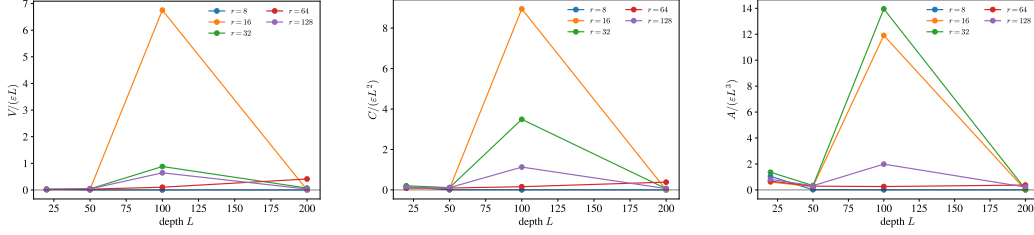


Figure 6: Deep finite-network cumulant stress test at depths $L \in \{20, 50, 100, 200\}$. The experiment is deliberately diagnostic: at very large depths the finite-network scalar-output cumulants become noisy and can show nonmonotone effective behavior, but the plots expose the regime where the polynomial normalization remains informative and where finite-depth effects dominate.

Table 1: Exact full-rank matching for true empirical NTKs at $n = r = 128$, $m = 8$, $d = 16$. Differences are at numerical precision, confirming Theorem 3.1.

Depth	max absolute NTK difference	relative difference
2	$8.88 \cdot 10^{-16}$	$8.09 \cdot 10^{-17}$
4	$3.55 \cdot 10^{-15}$	$1.31 \cdot 10^{-16}$
6	$5.33 \cdot 10^{-15}$	$4.05 \cdot 10^{-16}$
8	$3.55 \cdot 10^{-15}$	$4.20 \cdot 10^{-16}$
10	$6.22 \cdot 10^{-15}$	$1.14 \cdot 10^{-15}$

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A Mathematical Claim Ledger

The following ledger records the status of the mathematical ingredients used in the paper.

Status	Ingredients
Proved algebraically	Pathwise full-rank matching; fixed-left gradient decomposition; Haar covariance; rank cumulant power counting; endpoint moment calculation; three-layer delta-method expansion.
Conditional theorem/proposition	Tensor hierarchy; operator concentration; RF-LR spectral preservation; centered L^2 rule after radial projection.
Experimental diagnostic	Direct finite-network cumulant slopes and normalized ratios; operator proxy ratios; finite-depth effective constants outside isolated tensor coordinates.
Excluded from the claims	Strictly positive NMF treated as signed Gaussian low rank; training the left factor U while using fixed-projector diagrammatics.

B Projector Cumulants Beyond Second Order

For $G = (n/r)UU^\top$, the mean is $\mathbb{E}G = I_n$, the trace is deterministic, and every connected cumulant containing the trace direction vanishes. Orthogonal Weingarten calculus gives, for fixed cumulant order s ,

$$\text{cum}(G_{i_1 j_1}, \dots, G_{i_s j_s}) = O_s(r^{1-s}) \quad (27)$$

uniformly when $r \leq n$ and $r \rightarrow \infty$, with the full-rank endpoint cancelling after replacing r^{-1} by the defect scale $\gamma_{n,r}$ at second order. Diagrammatically, this is why a connected rank vertex is counted by

the number of connected projector insertions, not simply by the number of factors U in the algebraic expression.

C Low-Rank Diagrammatic Rules

This appendix records the graphical calculus underlying the perturbative statements in the main text. The external objects are the same as in finite-width NTK Feynman diagrams: NNGP/preactivation legs, NTK fluctuation legs, and derivative-NTK descendants. The difference is internal. In the low-rank model, stochastic vertices are empirical rank cumulants and frozen-projector cumulants rather than unconstrained neural-index contractions.

C.1 Visual dictionary

We use the same visual grammar as the finite-width NTK diagrammatics: filled external dots denote observed sample indices, solid lines denote NNGP/preactivation objects, colored dotted ghost lines denote NTK fluctuations, and multi-line ghost objects denote dNTK/ddNTK descendants. Low-rank only changes the internal stochastic vertices.

$$z_\alpha \equiv \alpha \bullet \text{---} \quad \widehat{\Delta\Theta}_{\alpha\beta} \equiv \begin{array}{c} \beta \bullet \text{---} \\ \alpha \bullet \text{---} \end{array} \quad \widehat{d\Theta} \equiv \begin{array}{c} \bullet \text{---} \\ \bullet \text{---} \\ \bullet \text{---} \end{array} \quad (28)$$

The new internal low-rank vertices are:

$$\text{rank cumulant} \equiv \text{---} \circ \text{---} \sim r^{1-s} \kappa_s, \quad \text{projector cumulant} \equiv \text{---} \diamond G \text{---} \sim \text{cum}(G, \dots, G). \quad (29)$$

C.2 Rank-state vertices

Let

$$M_A^{(\ell)} = \frac{1}{r_\ell} \sum_{a=1}^{r_\ell} \phi_A^{(\ell)}(Z_a), \quad (30)$$

where A labels a local atom such as $\sigma\sigma$, $\sigma\sigma'$, $\sigma'\sigma'$, or a derivative analogue. Conditional on the past, the rank atoms are iid. Therefore

$$\text{cum}(M_{A_1}^{(\ell)}, \dots, M_{A_s}^{(\ell)}) = r_\ell^{1-s} \kappa_{A_1 \dots A_s}^{(\ell)}, \quad \kappa_{A_1 \dots A_s}^{(\ell)} = \text{cum}(\phi_{A_1}^{(\ell)}(Z), \dots, \phi_{A_s}^{(\ell)}(Z)). \quad (31)$$

Thus every connected rank vertex of valence s has weight r_ℓ^{1-s} . In the isometric model, one also has frozen-projector vertices given by cumulants of $G = (n/r)UU^\top$. At valence two this is Lemma 4.1; at higher valence the weights are orthogonal Weingarten contractions. When $r = n$, $G = I$, so every centered projector vertex vanishes.

Theorem C.1 (Low-rank Feynman rules). *Let F be a smooth observable of finitely many layer states $M^{(\ell)}$ and frozen projectors $G^{(\ell)}$. Its expansion in powers of $1/r_\ell$ and projector cumulants is obtained by summing connected low-rank diagrams with the prescribed external legs. Algebraically:*

1. a rank vertex of valence s contributes $r_\ell^{1-s} \kappa_s^{(\ell)}$;
2. a projector vertex contributes the corresponding cumulant of $G^{(\ell)}$, with second-order scale γ_{n_ℓ, r_ℓ} ;
3. a deterministic layer vertex contributes a Frechet derivative $D^q \Phi_\ell$ of the layer map;
4. a transport edge contributes the product of first derivatives along the depth path;
5. for cumulants, disconnected diagrams cancel by the moment-cumulant relation.

Proof. Taylor expand F around the deterministic trajectory of the rank states. The cumulant generating operator for an empirical rank average is

$$\exp \left(\sum_{s \geq 2} \frac{r^{1-s}}{s!} \kappa_{A_1 \dots A_s} \partial_{A_1} \dots \partial_{A_s} \right), \quad (32)$$

up to the usual finite-order smoothness remainder. Expanding (32) gives the rank vertices and their deterministic derivative contractions. The same argument applies to G , replacing empirical-rank cumulants by frozen-projector cumulants. Taking connected cumulants removes disconnected products. $\square$

371 C.3 Basic tensor recursions

372 The diagrammatic rules produce the same external tensor hierarchy as the dense finite-width expansion, but with low-rank sources. Let J_ℓ^K and J_ℓ^Θ denote the deterministic linear transports for NNGP and NTK legs. The NNGP covariance block satisfies schematically

$$V_{\ell+1} = J_\ell^K V_\ell (J_\ell^K)^\top + \Xi_\ell^{KK}, \quad \Xi_\ell^{KK} = r_\ell^{-1} \kappa_2(S^K, S^K) + \gamma_{n_\ell, r_\ell} \mathfrak{h}_\ell^{KK}. \quad (33)$$

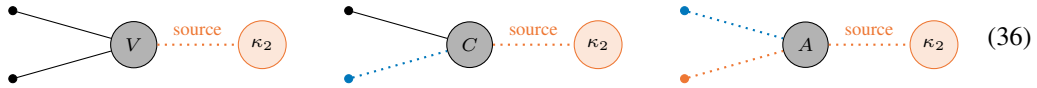
375 For the mixed NNGP-NTK block $C = (D, F)$,

$$C_{\ell+1} = J_\ell^K C_\ell (J_\ell^\Theta)^\top + S_V[V_\ell] + \Xi_\ell^{K\Theta}. \quad (34)$$

376 For the NTK-NTK block $A = (A, B)$,

$$A_{\ell+1} = J_\ell^\Theta A_\ell (J_\ell^\Theta)^\top + S_C[C_\ell] + S_V[V_\ell] + \Xi_\ell^{\Theta\Theta}. \quad (35)$$

377 The sources Ξ^{KK} , $\Xi^{K\Theta}$, and $\Xi^{\Theta\Theta}$ are precisely the rank and projector cumulant vertices with the corresponding external legs. This explains why the depth exponents in Proposition 5.2 come from external transport, while the small parameters come from r^{-1} , $1/n$, and $\gamma_{n,r}$.



380 C.4 First NTK mean correction

381 The familiar finite-width correction to the NTK mean has the same external five-term form:

$$\Theta_{\ell+1}^{\{1\}} = \mathsf{T}_\Theta[\Theta_\ell^{\{1\}}] + \mathsf{T}_K[K_\ell^{\{1\}}] + S_V[V_\ell] + S_D[D_\ell] + S_F[F_\ell]. \quad (37)$$

382 The low-rank difference is that the source tensors V, D, F are further resolved into the rank-state and projector vertices above. Thus the low-rank calculus does not change the external NTK grammar; it changes the internal expansion parameter from neural-index contractions to empirical-rank cumulants and frozen-projector cumulants.

386 C.5 Pairing extraction

387 A rank-four orthogonally invariant tensor decomposes as

$$C_{i_1 i_2 i_3 i_4} = c_1 \delta_{i_1 i_2} \delta_{i_3 i_4} + c_2 \delta_{i_1 i_3} \delta_{i_2 i_4} + c_3 \delta_{i_1 i_4} \delta_{i_2 i_3}. \quad (38)$$

388 Let B_1, B_2, B_3 be the three pairings. The normalized pairing Gram matrix is

$$\overline{G}_N = \begin{pmatrix} 1 & 1/N & 1/N \\ 1/N & 1 & 1/N \\ 1/N & 1/N & 1 \end{pmatrix}, \quad \overline{G}_N^{-1} = \frac{N}{N-1} I_3 - \frac{N}{(N-1)(N+2)} \mathbf{1}\mathbf{1}^\top. \quad (39)$$

389 If $h = (\langle C, B_1 \rangle, \langle C, B_2 \rangle, \langle C, B_3 \rangle)$ are measured contractions, the isolated tensor coefficients are
 390 $(c_1, c_2, c_3)^\top = \overline{G}_N^{-1} h$. This is why direct scalar-output cumulants should not be identified with the
 391 endpoint coefficients 5, 21/20, and 173/720 without a pairing-basis extraction.

392 C.6 Centered RF-LR memory lemma

393 For RF-LR, an old NTK line transported from layer k to L carries

$$B_{L:k} = \prod_{s=k+1}^L \frac{1}{r} \dot{\Sigma}^{(s)}. \quad (40)$$

At ReLU edge of chaos, $\dot{\Sigma}^{(s)} \rightarrow 1/2$, so $\|B_{L:k}\| \lesssim (2r)^{-(L-k)}$, up to polynomial endpoint factors. The corresponding centered transport diagram is

$$(41)$$

Let $P = I - \mathbf{1}\mathbf{1}^\top/m$ and $Z_L = P(\hat{K}_L - K_L)P$. The centered fluctuation admits the linearized expansion

$$Z_L = \sum_{k=1}^L B_{L:k+1} Y_k + \sum_{k=1}^L B_{L:k+1} R_k, \quad (42)$$

where Y_k is a centered fresh rank source and R_k is quadratic in earlier fluctuations.

Lemma C.2 (Centered RF-LR memory). *Assume, conditionally on the past,*

$$\|\mathbb{E}[Y_k^2 \mid \mathcal{F}_{k-1}]\|_{\text{op}} \leq C \frac{m\varepsilon_{\text{RF}}}{r^2}, \quad \|Y_k\|_{\psi_1} \leq C \frac{\sqrt{\varepsilon_{\text{RF}}}}{r}, \quad (43)$$

and $\|B_{L:k}\| \leq q^{L-k}$ for some $q < 1$. If $\|R_k\|_{\text{op}} \leq Cm\varepsilon_{\text{RF}}/r^2$ with high probability, then, up to logarithmic factors,

$$\|Z_L\|_{\text{op}} \lesssim \frac{\sqrt{m\varepsilon_{\text{RF}} \log(m/\delta)}}{r} + \frac{m\varepsilon_{\text{RF}} \log(m/\delta)}{r^2} \quad (44)$$

with probability at least $1 - \delta$.

Proof. Set $X_k = B_{L:k+1} Y_k$. The predictable matrix variance satisfies

$$\left\| \sum_{k=1}^L \mathbb{E}[X_k^2 \mid \mathcal{F}_{k-1}] \right\|_{\text{op}} \leq C \frac{m\varepsilon_{\text{RF}}}{r^2} \sum_{k=1}^L q^{2(L-k)} \leq C' \frac{m\varepsilon_{\text{RF}}}{r^2}. \quad (45)$$

Matrix Freedman or Bernstein gives the first term in (44). For the quadratic remainder,

$$\left\| \sum_{k=1}^L B_{L:k+1} R_k \right\|_{\text{op}} \leq C \frac{m\varepsilon_{\text{RF}}}{r^2} \sum_{k=1}^L q^{L-k} \leq C' \frac{m\varepsilon_{\text{RF}}}{r^2}. \quad (46)$$

□

Combining Lemma C.2 with Weyl's inequality gives the centered/angularized criterion. If the centered deterministic margin satisfies $\lambda_{\min}(PK_L P) \gtrsim (rL)^{-1}$, and if $\varepsilon_{\text{RF}} \simeq 1/r$, then the fluctuation bound is below the margin when

$$r \gtrsim mL^2 \quad (47)$$

up to logarithms. Without removing the radial constant mode, a diffusive $\sqrt{L}$ factor can reappear, giving the conservative cubic-depth criterion used for the unprojected RF-LR statement.

D Why Strict Positive NMF Is Not Covered

The model in (1) is signed and isometric. The right factor B is unconstrained and the frozen left factor U has orthonormal columns. Strict nonnegative matrix factorization imposes a cone constraint and changes the tangent space even at initialization. Its NTK is a projected or conic kernel, not the fixed linear-subspace kernel analyzed here. Therefore the results should not be cited as a theorem for positive NMF without a separate analysis of active constraints, boundary faces, and cone-projected gradients.

E Proof Details for Endpoint Constants

For $g \sim \mathcal{N}(0, 1)$, let $X_0 = 2(g_+)^2$ and $Y_0 = 2\mathbf{1}_{\{g>0\}}$. Since $\mathbb{E}[g_+^2] = 1/2$ and $\mathbb{E}[g_+^4] = 3/2$, we have $\mathbb{E}X_0 = 1$, $\mathbb{E}X_0^2 = 6$, and $\text{Var}(X_0) = 5$. Also $\mathbb{E}Y_0 = 1$, $\mathbb{E}Y_0^2 = 2$, so $\text{Var}(Y_0) = 1$. Finally, $\mathbb{E}X_0 Y_0 = 2$, hence $\text{Cov}(X_0, Y_0) = 1$. Transporting one or two NTK legs and applying the ReLU edge-of-chaos transport sum gives the formulas for B_λ and $A_{\lambda,\mu}$.

F Additional Experimental Figures

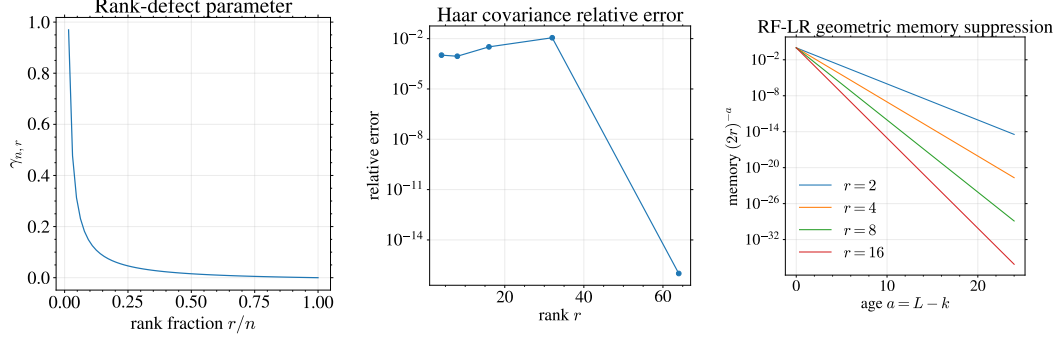


Figure 7: Additional proxy diagnostics: rank-defect parameter, relative Haar covariance error, and RF-LR geometric memory suppression.

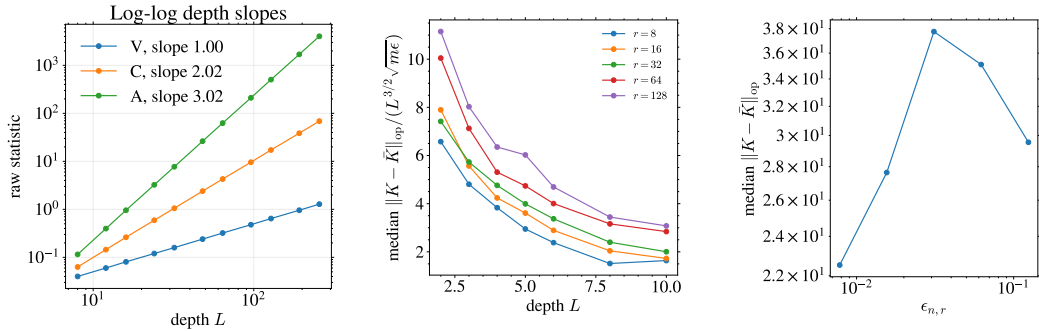


Figure 8: Depth-slope proxy diagnostics and true finite-network NTK operator/rank-defect checks.

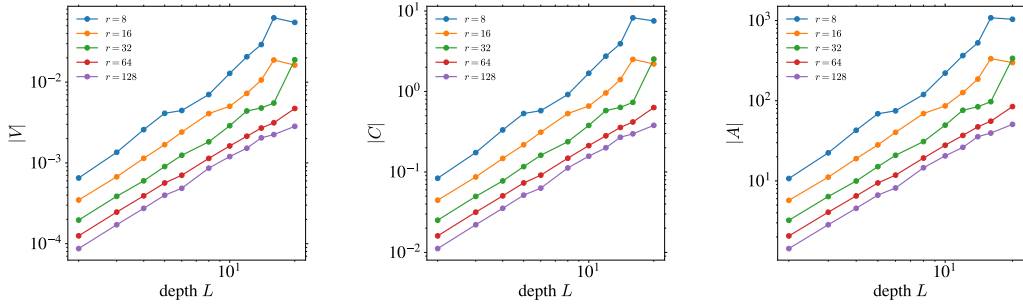


Figure 9: Log-log direct finite-network cumulant magnitudes. These plots complement the normalized-ratio plots in the main text.

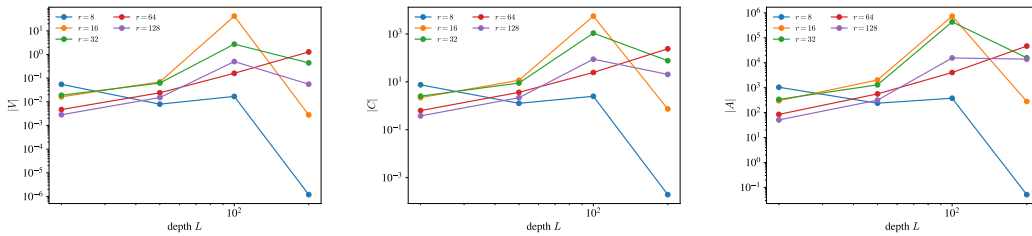


Figure 10: Deep-run log-log cumulant magnitudes for $L \in \{20, 50, 100, 200\}$. These plots are a stress test rather than a clean asymptotic fit: they reveal where finite-network cumulants become dominated by finite-depth and sampling effects.